

Who Pays the Bill?

Substrate Skepticism, Stable Counterfactual Identity,
Hidden Synergy, and the Causal Shannon Floor

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Abstract

Suppose an observable system is specified only by controlled stochastic input–output laws. Without identifying the underlying ontology, what information resources must an exact realization carry once one specifies which mutually exclusive possibilities are required to belong to one common latent construction?

The answer depends sharply on the realization contract. We distinguish chosen-path sampling, a one-step counterfactual table, a synchronous stable-event contract, and a general history-indexed causal response tree.

For the synchronous contract we retain the finite-horizon theory of the previous version. The minimum block entropy

$$C_n^{\text{syn}}$$

is subadditive, admits hidden cross-counterfactual synergy, and obeys an information-spectrum converse. For two actions P, Q ,

$$C_n^{\text{syn}}(P, Q) \geq \frac{n}{2} (\mathsf{H}(P) + \mathsf{H}(Q) + W_1(\mu_P, \mu_Q)),$$

where μ_P, μ_Q are the laws of one-symbol surprisal. For arbitrary finite action families the corresponding lower bound is a multi-marginal quantile envelope. For the equal-entropy three-outcome crash test,

$$1.728465468745531 \leq C_\infty^{\text{syn}} \leq 1.939147788079282.$$

The main new result concerns the weaker history-indexed causal contract. For every finite two-action memoryless controlled channel,

$$C_\infty^{\text{causal}}(P, Q) = \max\{\mathsf{H}(P), \mathsf{H}(Q)\}.$$

The proof is constructive. Long P - and Q -output streams are divided into blocks. Neighboring blocks are coupled almost at the larger Shannon entropy, while the Q -stream is reversed. The resulting dependence lies beyond a causal anti-diagonal: no horizon- n realized history can expose both sides of any coupled block. Thus the hidden dependence is operationally invisible on every admissible path while reducing the entropy of the complete causal seed.

For the crash-test pair,

$$C_\infty^{\text{causal}} = 1.5 < 1.728465468745531 \leq C_\infty^{\text{syn}}.$$

Hence stable synchronous cross-history event identity carries a strictly positive asymptotic information cost in this example, whereas unrestricted finite-horizon causal counterfactual coherence does not increase the asymptotic rate above the observable Shannon floor.

This conclusion concerns the finite-horizon block rate defined here. It does not by itself establish the existence of one projectively consistent infinite causal substrate with the same rate, and it does not identify the physical ontology of any realization.

The accountant discovered netting. The auditor discovered a lower bound. The lawyer asked which contract we had signed. The causal engineer moved the correlation behind the horizon.

1 Priority convention and philosophical origin

The purpose of this note is to sharpen a resource question, not to manufacture a novelty claim.

Priority convention. Assume that any clean statement derived here has probably been stated, proved, or embedded in a more general theory somewhere else. The word *derived* describes only the internal logic of this note. When a nearby precedent is known, it is cited. When no direct precedent has been located, we say only that we have not located one.

Minimum-entropy coupling is established information theory [2]. Information-spectrum converses for minimum-entropy coupling and functional representation are known [3]. Multi-marginal approximation, exact common information, exact channel synthesis, structured independence, marginal polytopes, causal transport, sample-privacy synergy, catalytic entropy, and finitary Bernoulli coding all provide neighboring mathematical mechanisms [4, 5, 6, 13, 14, 15, 16, 18, 19, 20, 21, 22].

The finite probability-space problem under prescribed independence constraints also has a long history; in particular Koller and Megiddo studied small sample spaces satisfying given probabilistic and independence constraints [26]. That literature concerns primarily support size and feasibility, whereas the objective below is Shannon entropy under a specific family of complete transversal constraints.

A philosophical trigger was Pavel Arazim’s public answer for the Czech outreach project *Zeptej se vědce* to the question “What if life is a dream and one wakes up after death?” [1]. The connection is motivational only and does not imply endorsement of the mathematics below.

The operational question used here is weaker:

If an exact realization exists under a stated contract, what resources must it carry?

We call this stance *substrate skepticism*. An implementation lower bound is not an ontological identification. Conversely, refusing ontology does not make exact realization resource-free.

2 The accounting-standard principle

A recurring failure mode in discussions of randomness, simulation, and counterfactuality is to move silently between inequivalent implementation contracts.

We distinguish four levels.

C1: Chosen-path sampling. At time t , the action A_t is already known. Only the requested output

$$Y_t \sim P_{A_t}$$

must be produced.

C2: One-step counterfactual table. Before the action is selected, one latent draw determines

$$F_t = (F_t(a))_{a \in \mathcal{A}}, \quad F_t(a) \sim P_a.$$

Only one coordinate is exposed, but all coordinates are jointly defined.

C3-S: Synchronous stable-event table. For horizon n , one latent source fixes

$$F = (F_t(a))_{1 \leq t \leq n, a \in \mathcal{A}}.$$

The coordinate indexed by the same (t, a) is required to denote the same latent event across all histories.

C3-C: History-indexed causal tree. A more general contingent plan fixes

$$F_t(h_{t-1}, a),$$

where the counterfactual output may depend on the complete prior history h_{t-1} .

Thus C3-S imposes a stronger cross-history identity than C3-C.

Accounting-standard principle.

The bill is meaningful only after the realization contract is fixed.

3 Controlled channels and the one-shot bill

Let $\mathcal{A}$ and $\mathcal{Y}$ be finite. A memoryless controlled channel is a family

$$W = \{P_a : a \in \mathcal{A}\}, \quad P_a \in \Delta(\mathcal{Y}),$$

satisfying

$$\Pr(Y_t = y \mid A_{1:t}, Y_{1:t-1}) = P_{A_t}(y) \quad (3.1)$$

under every adaptive action process.

Define

$$h_{\max} := \max_{a \in \mathcal{A}} H(P_a). \quad (3.2)$$

The policy that always selects an entropy-maximizing action produces $nh_{\max}$ bits of output entropy over n steps. Any exact latent source from which that path is a deterministic projection therefore has entropy at least $nh_{\max}$.

Under C2, the minimum entropy of a one-step latent table is

$$C_1(W) := \min \{H(F) : \text{Law}(F(a)) = P_a \ \forall a\}. \quad (3.3)$$

This is precisely the multi-marginal minimum-entropy coupling:

$$C_1(W) = \text{MEC}(P_a : a \in \mathcal{A}). \quad (3.4)$$

The one-shot surcharge is

$$r_1(W) := C_1(W) - h_{\max} \geq 0. \quad (3.5)$$

The central mistake to avoid is to multiply r_1 by time without proving that the chosen realization contract requires independent one-step tables.

4 Chosen-path sampling nearly reaches Shannon

Under C1 the counterfactual table is not required. Modern randomness-recycling methods show that exact online sampling can approach the Shannon cost while reusing unused randomness [8, 9]. Complementary results quantify the memory price of approaching this limit uniformly [10].

The relevant lesson here is structural:

adaptive choice alone does not force a positive entropy-rate surcharge.

Any additional bill must arise from a stronger requirement concerning which unrealized alternatives must remain jointly definite.

5 A strict reset obstruction

Before allowing hidden correlations, consider an intentionally strong reusable-reservoir model.

Let

$$S \sim \lambda, \quad U \sim \nu, \quad S \perp U,$$

where S and U have finite alphabets.

A strong reset recycler consists of deterministic maps

$$\Phi_a : S \times U \rightarrow \mathcal{Y} \times S$$

such that

$$\Phi_a(S, U) = (Y, S') \sim P_a \otimes \lambda \quad \forall a. \quad (5.1)$$

The returned reservoir therefore has its original law and is independent of the newly exposed output.

Proposition 5.1 (Entropy balance). *Every strong reset recycler satisfies*

$$H(\nu) \geq h_{\max}. \quad (5.2)$$

Proof. For every action a ,

$$H(P_a) + H(\lambda) = H(Y, S') \leq H(S, U) = H(\lambda) + H(\nu).$$

□

Theorem 5.2 (Exact reset floor forces spectral equivalence). *Let P, Q be finite probability vectors with*

$$H(P) = H(Q) = h.$$

If one finite strong reset recycler works for both actions with

$$H(\nu) = h,$$

then P and Q have the same multiset of nonzero atom probabilities, up to permutation and zero padding.

Proof. At equality, the deterministic map

$$(S, U) \mapsto (Y, S')$$

preserves entropy and is almost surely invertible on positive-probability atoms. Hence

$$\lambda \otimes \nu \equiv_{\text{perm}} P \otimes \lambda$$

and likewise

$$\lambda \otimes \nu \equiv_{\text{perm}} Q \otimes \lambda.$$

Thus

$$P \otimes \lambda \equiv_{\text{perm}} Q \otimes \lambda.$$

For every integer $r \geq 2$,

$$\left(\sum_i p_i^r \right) \left(\sum_j \lambda_j^r \right) = \left(\sum_i q_i^r \right) \left(\sum_j \lambda_j^r \right).$$

The second factor is positive, so

$$\sum_i p_i^r = \sum_i q_i^r \quad \forall r \geq 2.$$

After zero padding, Newton's identities determine the atom multiset. □

The theorem is a crash-test for one accounting standard. It is not a universal asymptotic rate theorem. Correlated catalytic constructions are precisely a warning that independence of a surviving auxiliary state can itself be a strong resource restriction [18, 19].

6 The equal-entropy three-outcome crash test

Throughout the concrete example, let

$$P = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right), \quad Q = (q, q, 1 - 2q), \quad (6.1)$$

where $q \in (1/3, 1/2)$ is the unique solution of

$$-2q \log_2 q - (1 - 2q) \log_2 (1 - 2q) = \frac{3}{2}. \quad (6.2)$$

Numerically,

$$q = 0.41003242818198557608 \dots \quad (6.3)$$

and

$$H(P) = H(Q) = 1.5. \quad (6.4)$$

The probability spectra are not permutations.

A vertex enumeration of the 3×3 transportation polytope gives a minimum-entropy coupling with nonzero atom probabilities

$$\frac{1}{2} - q, \quad q, \quad 2q - \frac{3}{4}, \quad 1 - 2q, \quad \frac{1}{4}. \quad (6.5)$$

Therefore

$$C_1(P, Q) = 2.0539185856072597 \dots \quad (6.6)$$

and

$$E_1 := C_1 - 1.5 = 0.5539185856072597 \dots \quad (6.7)$$

7 Synchronous stable-event blocks

For horizon n , let

$$F = (F_t(a))_{1 \leq t \leq n, a \in \mathcal{A}} \quad (7.1)$$

be one synchronous counterfactual table.

Definition 7.1 (Synchronous policy-safe law). *A law Γ of F is synchronous policy-safe if for every deterministic action string*

$$a^n = (a_1, \dots, a_n)$$

one has

$$\text{Law}_\Gamma(F_1(a_1), \dots, F_n(a_n)) = \bigotimes_{t=1}^n P_{a_t}. \quad (7.2)$$

Proposition 7.2 (Transversal criterion). *A synchronous table law is policy-safe if and only if it realizes the memoryless channel exactly for every deterministic nonanticipative policy. The same holds for randomized policies with private randomness independent of the table.*

Proof. Necessity follows from policies that ignore history.

Conversely, fix a deterministic policy and output string y^n . Along that branch, the policy determines an action string

$$a_t = a_t(y^{t-1}).$$

The branch event is

$$\bigcap_{t=1}^n \{F_t(a_t(y^{t-1})) = y_t\}.$$

By (??), its probability equals

$$\prod_{t=1}^n P_{a_t(y^{t-1})}(y_t),$$

which is exactly the controlled channel law. $\square$

Define

$$C_n^{\text{syn}}(W) := \min\{H(F) : F \text{ is synchronous policy-safe}\}. \quad (7.3)$$

The feasible set is a nonempty compact polytope.

Proposition 7.3 (Subadditivity and prefix lower bound). *For all $m, n \geq 1$,*

$$C_{m+n}^{\text{syn}} \leq C_m^{\text{syn}} + C_n^{\text{syn}}, \quad (7.4)$$

and

$$C_{m+n}^{\text{syn}} \geq mh_{\max} + C_n^{\text{syn}}. \quad (7.5)$$

Consequently

$$C_{\infty}^{\text{syn}} := \lim_{n \rightarrow \infty} \frac{C_n^{\text{syn}}}{n} = \inf_{n \geq 1} \frac{C_n^{\text{syn}}}{n}. \quad (7.6)$$

Proof. Independent concatenation gives (??).

For (??), select an entropy-maximizing action during the first m times and retain the entire future n -step table. Conditioned on every positive-probability realized prefix, the future table remains synchronous policy-safe. Hence the corresponding deterministic projection of the full table has entropy at least

$$mh_{\max} + C_n^{\text{syn}}.$$

Fekete's lemma gives (??). $\square$

For equal-entropy actions define

$$E_n^{\text{syn}} := C_n^{\text{syn}} - nh. \quad (7.7)$$

Then E_n^{syn} is nondecreasing.

7.1 The algebraic skeleton of the transversal polytope

Suppose action a has output alphabet size d_a . At one time the complete counterfactual state lies in

$$R := \prod_{a \in \mathcal{A}} \mathcal{Y}_a.$$

Let V_a be the vector space of real functions on R that depend only on coordinate a , and define

$$\mathcal{O} := \sum_{a \in \mathcal{A}} V_a. \quad (7.8)$$

Proposition 7.4 (Observable-space dimension and block rank). *The one-time observable space has dimension*

$$d_{\text{obs}} = 1 + \sum_{a \in \mathcal{A}} (d_a - 1). \quad (7.9)$$

The row space generated by all n -time transversal marginal constraints is

$$\mathcal{O}^{\otimes n}. \quad (7.10)$$

Hence

$$\text{rank } A_n = d_{\text{obs}}^n. \quad (7.11)$$

For two ternary actions,

$$|R| = 9, \quad d_{\text{obs}} = 5,$$

and therefore

$$\text{rank } A_n = 5^n, \quad \dim \ker A_n = 9^n - 5^n. \quad (7.12)$$

The kernel contains the linear directions invisible to every allowed transversal. This is the natural linear home of hidden counterfactual interaction.

8 Hidden synergy

The synchronous constraints do not force full one-time tables at different times to be independent.

Let γ be any one-time coupling of the required action marginals on the complete one-time state R .

Assume there exists a bounded nonzero function $h : R \rightarrow \mathbb{R}$ such that

$$\mathbb{E}_\gamma h = 0, \quad \mathbb{E}_\gamma [h \mid R(a)] = 0 \quad \forall a. \quad (8.1)$$

Proposition 8.1 (Hidden-interaction perturbation). *For sufficiently small nonzero ε ,*

$$\Gamma_\varepsilon(r, s) = \gamma(r)\gamma(s) (1 + \varepsilon h(r)h(s)) \quad (8.2)$$

is a two-step synchronous policy-safe law. Its one-time marginals are both γ , while

$$I(R_1; R_2) > 0.$$

Consequently

$$H(R_1, R_2) < 2H(\gamma). \quad (8.3)$$

Proof. For sufficiently small $|\varepsilon|$, nonnegativity holds. When unobserved coordinates are summed out, the perturbation term factorizes through conditional expectations in (??) and vanishes. Thus every selectable cross-time coordinate pair remains independent with the prescribed marginals.

The joint law differs from $\gamma \otimes \gamma$, so the temporal mutual information is positive. $\square$

This explains why hidden interaction is generic whenever the one-time coupling admits a nontrivial pure interaction direction. It does not by itself prove a saving relative to $2C_1$, because the chosen γ need not minimize one-step entropy.

8.1 An explicit 23-atom two-step witness

For the crash-test pair, let

$$R_t = (F_t(P), F_t(Q)) \in \{0, 1, 2\}^2.$$

Appendix A gives an explicit sparse law $\tilde{G}$ on

$$(x_1, z_1, x_2, z_2) \in \{0, 1, 2\}^4.$$

Proposition 8.2 (Explicit hidden-synergy witness). *At the value of q in (??), $\tilde{G}$ has exactly 23 positive atoms and satisfies*

$$(F_1(a), F_2(b)) \sim P_a \otimes P_b \quad \forall a, b \in \{P, Q\}. \quad (8.4)$$

Hence it is synchronous policy-safe.

Numerically,

$$H(\tilde{G}) = 3.878295576158564 \dots \quad (8.5)$$

For this witness,

$$H(R_1) = 2.860191813676764 \dots, \quad (8.6)$$

$$H(R_2) = 2.841039077264926 \dots, \quad (8.7)$$

and

$$I(R_1; R_2) = 1.822935314783125 \dots \quad (8.8)$$

More strikingly,

$$I(F_1(P); R_2) = 0.617663017691050 \dots, \quad (8.9)$$

while

$$I(F_1(P); F_2(P)) = I(F_1(P); F_2(Q)) = 0. \quad (8.10)$$

Thus a realized past can carry information about the *joint* future counterfactual table while carrying no information about either individually selectable future coordinate.

Corollary 8.3 (Strict synchronous block subadditivity). *For the crash-test pair,*

$$C_2^{\text{syn}} \leq 3.878295576158564 < 2C_1. \quad (8.11)$$

The certified saving is

$$2C_1 - H(\tilde{G}) = 0.229541595055955 \dots \quad (8.12)$$

Hence

$$C_\infty^{\text{syn}} \leq 1.939147788079282 \dots \quad (8.13)$$

8.2 A variational identity for net hidden synergy

For a synchronous policy-safe law let

$$R_t = (F_t(a))_{a \in \mathcal{A}}$$

be the complete one-time table and define temporal total correlation

$$\text{TC}(R_{1:n}) := \sum_{t=1}^n H(R_t) - H(R_{1:n}). \quad (8.14)$$

Since

$$H(R_t) \geq C_1,$$

we have

$$H(R_{1:n}) = nC_1 + \sum_{t=1}^n (H(R_t) - C_1) - \text{TC}(R_{1:n}). \quad (8.15)$$

Taking the minimum gives

$$\boxed{nC_1 - C_n^{\text{syn}} = \max_{\Gamma \text{ policy-safe}} \left[\text{TC}_{\Gamma}(R_{1:n}) - \sum_{t=1}^n (H_{\Gamma}(R_t) - C_1) \right].} \quad (8.16)$$

Net hidden synergy is therefore temporal dependence minus the additional one-time entropy required to make that dependence possible.

9 The reciprocal-scale overlap kernel

An earlier author manuscript considered

$$\rho_k(x) = \frac{k+1}{2}(1-|x|)_+^k, \quad \rho_{k,a}(x) = \frac{1}{a}\rho_k(x/a), \quad (9.1)$$

and

$$F_k(a, b) = \int_{\mathbb{R}} \rho_{k,a}(x) \rho_{k,b}(x) dx. \quad (9.2)$$

For $0 < a \leq b$,

$$F_k(a, b) = \frac{1}{b} H_k(a/b) \quad (9.3)$$

for an explicit polynomial/hypergeometric function H_k [24].

Only the elementary $k = 0$ case is needed here. Since $\rho_{0,a}$ is uniform on $[-a, a]$,

$$F_0(a, b) = \frac{1}{2 \max\{a, b\}}. \quad (9.4)$$

Under

$$a = \frac{1}{p}, \quad b = \frac{1}{q},$$

one obtains the exact identity

$$\boxed{2F_0(1/p, 1/q) = \min\{p, q\}}. \quad (9.5)$$

Writing

$$u = -\log_2 p, \quad v = -\log_2 q,$$

gives

$$2F_0(2^u, 2^v) = 2^{-\max\{u, v\}}, \quad (9.6)$$

and hence

$$\boxed{-\log_2(2F_0(2^u, 2^v)) = \max\{u, v\}}. \quad (9.7)$$

This is an identity, not an analogy. The earlier kernel calculation did not contain the counterfactual problem. It merely contains an exact overlap representation of the scalar minimum that later appears in the information-spectrum converse.

For general k ,

$$F_k(1/p, 1/q) = \min(p, q) H_k\left(\frac{\min(p, q)}{\max(p, q)}\right), \quad (9.8)$$

but no stronger counterfactual theorem is claimed from the higher- k family.

10 Synchronous information-spectrum geometry

For a finite distribution $P = (p_i)$, define the surprisal variable

$$\iota_P(X) := -\log_2 P(X), \quad X \sim P, \quad (10.1)$$

and its law

$$\mu_P := \text{Law}(\iota_P(X)). \quad (10.2)$$

10.1 Two actions

Let

$$X_t := F_t(P), \quad Z_t := F_t(Q).$$

Theorem 10.1 (Two-action synchronous information-spectrum bound). *Every synchronous policy-safe n -block satisfies*

$$\boxed{\mathbf{H}(F) \geq \frac{n}{2} (\mathbf{H}(P) + \mathbf{H}(Q) + W_1(\mu_P, \mu_Q))}. \quad (10.3)$$

Consequently

$$\boxed{C_\infty^{\text{syn}}(P, Q) \geq \frac{1}{2} (\mathbf{H}(P) + \mathbf{H}(Q) + W_1(\mu_P, \mu_Q))}. \quad (10.4)$$

Proof. Fix a positive-probability full-table atom

$$f = (x_1, z_1, \dots, x_n, z_n)$$

with probability $\Gamma(f)$.

For every subset $S \subseteq [n]$, the same atom lies inside the transversal event selecting P on S and Q on S^c . Therefore

$$\Gamma(f) \leq \prod_{t \in S} P(x_t) \prod_{t \notin S} Q(z_t). \quad (10.5)$$

Minimizing over S separates coordinatewise:

$$\Gamma(f) \leq \prod_{t=1}^n \min\{P(x_t), Q(z_t)\}. \quad (10.6)$$

Using (??), this is equivalently a product of reciprocal-scale $k = 0$ overlap kernels. Taking $-\log_2$,

$$-\log_2 \Gamma(f) \geq \sum_{t=1}^n \max\{\iota_P(x_t), \iota_Q(z_t)\}. \quad (10.7)$$

Average over F . At each time, the pair of surprisals has some coupling of μ_P, μ_Q . Since

$$\max(u, v) = \frac{1}{2} (u + v + |u - v|),$$

minimizing the expected absolute difference gives the one-dimensional Wasserstein distance. $\square$

If

$$\mathbf{H}(P) = \mathbf{H}(Q) = h,$$

then

$$\boxed{C_\infty^{\text{syn}}(P, Q) \geq h + \frac{1}{2} W_1(\mu_P, \mu_Q)}. \quad (10.8)$$

Moreover,

$$W_1(\mu_P, \mu_Q) = 0$$

if and only if the nonzero atom spectra of P, Q agree up to permutation. Thus unequal probability spectra with equal Shannon entropy create a strict synchronous rate surcharge.

10.2 Arbitrary finite action families

For each action $a \in \mathcal{A}$, let

$$U_a := -\log_2 P_a(Y_a), \quad Y_a \sim P_a,$$

and let

$$\mu_a := \text{Law}(U_a).$$

Define

$$\mathcal{T}(P_a : a \in \mathcal{A}) := \inf_{\eta \in \Pi(\mu_a : a \in \mathcal{A})} \mathbb{E}_\eta \max_{a \in \mathcal{A}} U_a. \quad (10.9)$$

Lemma 10.2 (Quantile envelope). *For finite-support laws μ_a ,*

$$\boxed{\mathcal{T}(P_a : a \in \mathcal{A}) = \int_0^1 \max_{a \in \mathcal{A}} F_{\mu_a}^{-1}(u) du.} \quad (10.10)$$

Proof. For every coupling,

$$\Pr\left(\max_a U_a \leq t\right) = \Pr(U_a \leq t \ \forall a) \leq \min_a F_{\mu_a}(t).$$

The comonotone coupling

$$U_a = F_{\mu_a}^{-1}(V), \quad V \sim \text{Unif}[0, 1],$$

attains equality simultaneously. □

Theorem 10.3 (Multi-action synchronous information-spectrum bound). *For every finite action family,*

$$\boxed{C_n^{\text{syn}}(P_a : a \in \mathcal{A}) \geq n \int_0^1 \max_{a \in \mathcal{A}} F_{\mu_a}^{-1}(u) du.} \quad (10.11)$$

Proof. A full-table atom must fit inside every action-string transversal. Hence

$$\Gamma(f) \leq \prod_{t=1}^n \min_{a \in \mathcal{A}} P_a(f_{t,a}), \quad (10.12)$$

so

$$-\log_2 \Gamma(f) \geq \sum_{t=1}^n \max_{a \in \mathcal{A}} [-\log_2 P_a(f_{t,a})].$$

Average and apply the quantile-envelope lemma. □

11 The synchronous crash-test interval

For

$$P = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right),$$

the surprisal law is

$$\mu_P = \frac{1}{2}\delta_1 + \frac{1}{2}\delta_2. \quad (11.1)$$

For

$$\begin{aligned} Q &= (q, q, 1 - 2q), \\ \mu_Q &= 2q \delta_{-\log_2 q} + (1 - 2q) \delta_{-\log_2(1-2q)}. \end{aligned} \quad (11.2)$$

Numerically,

$$W_1(\mu_P, \mu_Q) = 0.4569309374910615 \dots \quad (11.3)$$

Therefore

$$C_\infty^{\text{syn}} \geq 1.5 + \frac{1}{2}W_1 = 1.7284654687455308 \dots \quad (11.4)$$

Together with the 23-atom witness,

$$\boxed{1.728465468745531 \leq C_\infty^{\text{syn}} \leq 1.939147788079282.} \quad (11.5)$$

At horizon two,

$$3.553918585607260 \leq C_2^{\text{syn}} \leq 3.878295576158564. \quad (11.6)$$

The upper endpoint is a feasible witness value, not a certified global minimum.

12 Counterfactual breadth

Full synchronous policy-safety requires every action string.

For two actions, let

$$\mathcal{G} \subseteq \{P, Q\}^n$$

be only the family of transversals required to be correct, and let $C_n(\mathcal{G})$ denote the corresponding minimum full-table entropy.

For a full atom define

$$M_t := \max\{\imath_P(X_t), \imath_Q(Z_t)\},$$

$$\Delta_t := |\imath_P(X_t) - \imath_Q(Z_t)|.$$

Let $s \in \{P, Q\}^n$ choose at each coordinate the locally more surprising action. For $g \in \mathcal{G}$, define the weighted Hamming distance

$$d_\Delta(s, g) := \sum_{t: s_t \neq g_t} \Delta_t. \quad (12.1)$$

Then the exact pointwise bound is

$$-\log_2 \Gamma(f) \geq \sum_t M_t - d_\Delta(s, \mathcal{G}), \quad (12.2)$$

where

$$d_\Delta(s, \mathcal{G}) = \min_{g \in \mathcal{G}} d_\Delta(s, g).$$

Assume positive support and define

$$D := \max_{x, z} |\imath_P(x) - \imath_Q(z)|. \quad (12.3)$$

Let

$$r(\mathcal{G}) = \max_s \min_{g \in \mathcal{G}} d_H(s, g) \quad (12.4)$$

be the ordinary Hamming covering radius.

Then

$$C_n(\mathcal{G}) \geq n \mathcal{T}(P, Q) - D r(\mathcal{G}). \quad (12.5)$$

Thus counterfactual breadth is a covering geometry, not merely a count. If

$$r(\mathcal{G}_n) = o(n),$$

the full synchronous information-spectrum rate survives asymptotically.

13 From synchronous block entropy to innovation rate

The quantity C_n^{syn} prices a precommitted source that fixes the complete stable-event table. It must not be silently identified with every possible notion of dynamic fresh randomness.

Suppose

$$F^{(n)} = \phi_n(S_0, U_{1:n}), \quad (13.1)$$

where S_0 is an initial reservoir independent of the external innovation source.

Then

$$C_n^{\text{syn}} \leq H(F^{(n)}) \leq H(S_0) + H(U_{1:n}),$$

so

$$\boxed{H(U_{1:n}) \geq C_n^{\text{syn}} - H(S_0)}. \quad (13.2)$$

If

$$H(S_0) = o(n),$$

then

$$\liminf_{n \rightarrow \infty} \frac{1}{n} H(U_{1:n}) \geq C_\infty^{\text{syn}}. \quad (13.3)$$

For an entropy-priced precommitted source, the opposite inequality is obtained by drawing an optimal block table and concatenating blocks. Under this explicit accounting standard,

$$R_{\text{precommitted}} = C_\infty^{\text{syn}}. \quad (13.4)$$

14 History-indexed causal trees

The synchronous table identifies $F_t(a)$ across all histories. The general causal contract instead permits

$$F_t(h_{t-1}, a).$$

Definition 14.1 (Causal-tree entropy). *Let*

$$C_n^{\text{causal}}(W)$$

be the minimum Shannon entropy of a finite random deterministic history-indexed response tree whose induced realized process is exactly the memoryless controlled channel under every adaptive nonanticipative policy.

At horizon one,

$$C_1^{\text{causal}} = C_1. \quad (14.1)$$

Every synchronous table induces a causal tree by defining

$$F_t(h, a) = F_t(a),$$

so

$$\boxed{C_n^{\text{causal}} \leq C_n^{\text{syn}}}. \quad (14.2)$$

14.1 The action-tree quotient

A deterministic history-indexed tree contains labels at histories that may never be reached by that deterministic seed. Those labels need not be retained in a minimum behavioral description.

For every action word

$$a^t = (a_1, \dots, a_t) \in \mathcal{A}^t,$$

let

$$Z_{a^t}$$

be the output generated at depth t when the action word a^t is fed to the deterministic seed.

Thus one obtains a random action-labelled tree

$$Z = (Z_v)_{v \in \mathcal{A}^{\leq n}}.$$

Proposition 14.2 (Action-tree characterization). *A random action tree is an exact horizon- n causal realization if and only if*

$$\text{Law}(Z_{a_1}, Z_{a_1 a_2}, \dots, Z_{a_1 \dots a_n}) = \bigotimes_{t=1}^n P_{a_t} \quad (14.3)$$

for every deterministic action word $a^n \in \mathcal{A}^n$.

Consequently,

$$\boxed{C_n^{\text{causal}} = \min H((Z_v)_{v \in \mathcal{A}^{\leq n}})}, \quad (14.4)$$

where the minimum is over action trees satisfying (??).

Proof. Every history-indexed deterministic tree induces such an action tree.

Conversely, fix an adaptive deterministic policy and an output string y^n . That output string recursively determines one deterministic action word

$$a_t = a_t(y^{t-1}).$$

The event that the policy produces y^n is exactly the corresponding action-tree path event. Equation (??) therefore supplies the required channel probability. $\square$

14.2 Causal prefix recursion

Proposition 14.3 (Causal prefix lower bound). *For every $n \geq 1$,*

$$C_n^{\text{causal}} \geq C_1 + (n-1)h_{\max}. \quad (14.5)$$

Moreover,

$$C_{m+n}^{\text{causal}} \leq C_m^{\text{causal}} + C_n^{\text{causal}}. \quad (14.6)$$

Hence

$$C_{\infty}^{\text{causal}} := \lim_{n \rightarrow \infty} \frac{C_n^{\text{causal}}}{n} \quad (14.7)$$

exists and

$$h_{\max} \leq C_{\infty}^{\text{causal}} \leq C_{\infty}^{\text{syn}}. \quad (14.8)$$

Proof. Choose an entropy-maximizing root action. Its root output Y and the realized child subtree T_Y are a deterministic function of the complete seed. Thus

$$H(S_n) \geq H(Y, T_Y) = h_{\max} + H(T_Y | Y).$$

Conditioned on every positive-probability root output, the child subtree must again realize the same controlled channel for every continuation policy. Therefore

$$C_n^{\text{causal}} \geq h_{\max} + C_{n-1}^{\text{causal}}.$$

Iterate.

Independent concatenation of causal blocks gives subadditivity. $\square$

14.3 Residual hidden-tree entropy

Assume now that every action has the same Shannon entropy

$$H(P_a) = h.$$

Define

$$E_n := C_n^{\text{causal}} - nh. \quad (14.9)$$

Fix any deterministic policy π . For every exact causal seed S_n , the realized trajectory Y_π^n is a deterministic function of S_n . Moreover,

$$H(Y_\pi^n) = \sum_{t=1}^n H(Y_t \mid Y_\pi^{t-1}) = nh.$$

Therefore

$$\boxed{H(S_n) = nh + H(S_n \mid Y_\pi^n).} \quad (14.10)$$

Minimizing over exact seeds yields

$$\boxed{E_n = \inf_{S_n} H(S_n \mid Y_\pi^n).} \quad (14.11)$$

Thus E_n is exactly the hidden entropy remaining after one complete realized history is revealed.

The causal prefix bound gives

$$E_{n+1} \geq E_n, \quad (14.12)$$

while concatenation gives

$$E_{m+n} \leq E_m + E_n. \quad (14.13)$$

Hence E_n is nonnegative, nondecreasing, and subadditive.

Furthermore,

$$\boxed{C_\infty^{\text{causal}} - h = \lim_{n \rightarrow \infty} \frac{E_n}{n}.} \quad (14.14)$$

The causal asymptotic problem is therefore exactly the question

$$E_n = o(n) \quad \text{versus} \quad E_n = \Omega(n). \quad (14.15)$$

15 An exact causal Bellman–MEC recursion

Let $\mathcal{M}_n$ be the set of laws of exact depth- n action trees.

At the root, for action a , write

$$B_a = (Y_a, T_a),$$

where T_a is the continuation tree below the selected root branch.

Conditional on

$$Y_a = y,$$

let

$$K_{a,y} := \text{Law}(T_a \mid Y_a = y).$$

Exactness implies

$$K_{a,y} \in \mathcal{M}_n.$$

The marginal law of the complete a -branch is the disjoint mixture

$$M_a = \bigoplus_y P_a(y) K_{a,y}. \quad (15.1)$$

Theorem 15.1 (Exact causal Bellman–MEC recursion). *For every $n \geq 1$,*

$$\boxed{C_{n+1}^{\text{causal}} = \inf_{\{K_{a,y} \in \mathcal{M}_n\}} \text{MEC}(M_a : a \in \mathcal{A}).} \quad (15.2)$$

Proof. Once all conditional continuation laws $K_{a,y}$ are fixed, each root action has prescribed branch marginal M_a .

Branches belonging to distinct root actions occupy disjoint parts of the action tree. They may therefore be joined by an arbitrary common coupling. The minimum entropy of that coupling is exactly their multi-marginal MEC. Finally minimize over the continuation laws. $\square$

This recursion shows why a rigid iteration

$$R_{n+1} = \text{MEC}(P \otimes R_n, Q \otimes R_n)$$

is not the full causal optimization problem. Causality allows continuation laws to depend on both the root action and the root outcome.

15.1 Shape cost versus joining cost

For equal-entropy actions define

$$e_a = \sum_y P_a(y) \left[H(K_{a,y}) - C_n^{\text{causal}} \right] \geq 0. \quad (15.3)$$

Then

$$H(M_a) = h + C_n^{\text{causal}} + e_a. \quad (15.4)$$

Define

$$\Delta_{\text{shape}} := \max_a e_a, \quad (15.5)$$

and

$$\Delta_{\text{join}} := \text{MEC}(M_a : a \in \mathcal{A}) - \max_a H(M_a). \quad (15.6)$$

Both quantities are nonnegative, and

$$\boxed{E_{n+1} - E_n = \inf_K (\Delta_{\text{shape}} + \Delta_{\text{join}}).} \quad (15.7)$$

Thus a causal realization can reduce its next joining cost only by choosing continuation spectra that become easier to couple, possibly at the price of moving away from entropy-optimal child trees.

This is the precise form of the *shape-versus-joining* tradeoff.

16 A finite-depth rigidity result

The eventual causal rate will turn out to be the Shannon floor. Nevertheless, finite horizons do not attain that floor for the crash test.

The one-step transportation polytope has 18 vertices. Up to permutation, their nonzero atom spectra fall into five types. The minimum-entropy type is

$$\lambda_{\star} = \left(q, \frac{1}{4}, 1 - 2q, \frac{1}{2} - q, 2q - \frac{3}{4} \right). \quad (16.1)$$

On

$$\frac{3}{8} < q < \frac{5}{12}, \quad (16.2)$$

this spectrum strictly majorizes all four competing vertex-spectrum types. Since Shannon entropy is strictly Schur-concave, every one-step MEC minimizer has the same nonzero atom multiset $\lambda_{\star}$.

Theorem 16.1 (Strict growth at depth two). *For the crash-test pair,*

$$\boxed{C_2^{\text{causal}} > h + C_1.} \quad (16.3)$$

Equivalently,

$$\boxed{E_2 > E_1.} \quad (16.4)$$

Proof. Assume instead

$$C_2^{\text{causal}} = h + C_1.$$

For root action a , let

$$B_a = (Y_a, T_a).$$

Every conditional child law

$$K_{a,y} = \text{Law}(T_a \mid Y_a = y)$$

is a valid one-step counterfactual coupling and hence

$$H(K_{a,y}) \geq C_1.$$

Therefore

$$H(B_a) = h + \sum_y P_a(y) H(K_{a,y}) \geq h + C_1.$$

Since B_a is a deterministic function of the full depth-two seed, equality of the global entropy forces equality everywhere. Thus every $K_{a,y}$ is a one-step MEC minimizer and has spectrum $\lambda_\star$.

Consequently,

$$\text{spec}(B_P) = \text{spec}(P \otimes \lambda_\star),$$

and

$$\text{spec}(B_Q) = \text{spec}(Q \otimes \lambda_\star).$$

Both branch objects have the same entropy as the global seed and are deterministic images of it. They must therefore have the same atom multiset. Hence

$$P \otimes \lambda_\star \equiv_{\text{perm}} Q \otimes \lambda_\star.$$

Power-sum cancellation then gives

$$P \equiv_{\text{perm}} Q,$$

contradicting the crash-test construction. □

Since E_n is nondecreasing and

$$E_1 > 0,$$

no finite causal horizon reaches exactly

$$nh.$$

The next sections show that this finite residual is nevertheless sublinear.

17 Low-entropy couplings of long product blocks

The causal asymptotic construction requires one ingredient: long P - and Q -blocks can be coupled with entropy equal to the larger Shannon entropy plus a sublinear correction.

For a finite probability vector A , let

$$U_A = -\log_2 A(X), \quad X \sim A,$$

and define

$$V_A := \text{Var}(U_A). \quad (17.1)$$

For a pair A, B , Compton et al. define a profile lower envelope for minimum-entropy coupling and prove that the standard greedy coupling $G(A, B)$ satisfies

$$\text{H}(G(A, B)) \leq \text{H}(\text{PROFILE}(A, B)) + c_0, \quad (17.2)$$

where

$$c_0 = \frac{\log_2 e}{e} = 0.5307 \dots \quad (17.3)$$

[25].

For two distributions the profile entropy has an equivalent surprisal representation.

Lemma 17.1 (Profile entropy as surprisal transport). *For finite distributions A, B ,*

$$\boxed{\text{H}(\text{PROFILE}(A, B)) = \frac{1}{2} [\text{H}(A) + \text{H}(B) + W_1(\mu_A, \mu_B)]}. \quad (17.4)$$

Proof. In the mass-coordinate representation of the profile, taking $-\log_2$ converts the lower envelope of atom sizes into the upper envelope of the corresponding surprisal quantiles. Hence

$$\text{H}(\text{PROFILE}(A, B)) = \int_0^1 \max \{F_{\mu_A}^{-1}(u), F_{\mu_B}^{-1}(u)\} du.$$

Using

$$\max(x, y) = \frac{1}{2}(x + y + |x - y|)$$

and the one-dimensional quantile identity

$$W_1(\mu_A, \mu_B) = \int_0^1 |F_{\mu_A}^{-1}(u) - F_{\mu_B}^{-1}(u)| du$$

gives the result. □

Combining (??) and (??) yields the following useful existence statement.

Lemma 17.2 (Two-block coupling bound). *Let P, Q be finite distributions and define*

$$h_P = \text{H}(P), \quad h_Q = \text{H}(Q), \quad h_\star = \max\{h_P, h_Q\}.$$

Let

$$V_P = \text{Var}(\iota_P(X)), \quad V_Q = \text{Var}(\iota_Q(Z)).$$

For every integer $L \geq 1$, there exists a coupling

$$J_L \in \Pi(P^{\otimes L}, Q^{\otimes L})$$

such that

$$\boxed{\text{H}(J_L) \leq Lh_\star + A\sqrt{L} + c_0}, \quad (17.5)$$

where

$$A = \frac{1}{2} \left(\sqrt{V_P} + \sqrt{V_Q} \right). \quad (17.6)$$

Proof. The surprisal under $P^{\otimes L}$ is

$$S_P^{(L)} = \sum_{i=1}^L U_i,$$

with

$$\mathbb{E}S_P^{(L)} = Lh_P, \quad \text{Var}(S_P^{(L)}) = LV_P.$$

Likewise for Q .

Choose any coupling of the centered sums. By the triangle inequality and Cauchy–Schwarz,

$$W_1(\mu_{P^{\otimes L}}, \mu_{Q^{\otimes L}}) \leq L|h_P - h_Q| + \sqrt{LV_P} + \sqrt{LV_Q}. \quad (17.7)$$

Therefore the profile identity gives

$$\begin{aligned} \mathsf{H}(\text{PROFILE}(P^L, Q^L)) &\leq \frac{1}{2} \left[Lh_P + Lh_Q + L|h_P - h_Q| \right. \\ &\quad \left. + \sqrt{LV_P} + \sqrt{LV_Q} \right] \\ &= Lh_\star + A\sqrt{L}. \end{aligned}$$

Apply the greedy-profile bound (??). □

The only feature needed below is

$$\boxed{\mathsf{H}(J_L) = Lh_\star + o(L).} \quad (17.8)$$

18 The anti-diagonal causal construction

We now turn to the central construction.

Let the channel have exactly two actions, called P and Q .

Fix integers

$$m \geq 2, \quad L \geq 1, \quad n = mL.$$

Introduce two length- n streams

$$X_1, \dots, X_n$$

and

$$W_1, \dots, W_n.$$

Divide them into m consecutive blocks:

$$X^{(1)}, \dots, X^{(m)}, \quad W^{(1)}, \dots, W^{(m)},$$

each of length L .

Let J_L be any coupling of

$$Q^{\otimes L} \quad \text{and} \quad P^{\otimes L}.$$

Construct the source as follows:

1. draw

$$X^{(1)} \sim P^{\otimes L};$$

2. draw

$$W^{(m)} \sim Q^{\otimes L};$$

3. for every

$$j = 1, \dots, m - 1,$$

draw independently

$$(W^{(j)}, X^{(j+1)}) \sim J_L;$$

4. make all $m + 1$ objects above mutually independent.

Finally reverse the W -stream:

$$Z_k := W_{n+1-k}, \quad 1 \leq k \leq n. \quad (18.1)$$

The deterministic causal response rule is:

- on the r -th occurrence of action P , output X_r ;
- on the s -th occurrence of action Q , output Z_s .

18.1 Why the hidden correlations are causally invisible

The only dependence in the construction is between

$$W^{(j)} \quad \text{and} \quad X^{(j+1)}. \quad (18.2)$$

A horizon- n action history that has used i copies of action P has observed only

$$X_1, \dots, X_i. \quad (18.3)$$

If it has used k copies of action Q , then because of the reversal it has observed only the suffix

$$W_n, W_{n-1}, \dots, W_{n-k+1}. \quad (18.4)$$

Lemma 18.1 (Anti-diagonal invisibility). *No action history of length at most n can observe both sides of the same coupled pair*

$$(W^{(j)}, X^{(j+1)}).$$

Proof. To observe any coordinate of $X^{(j+1)}$, the history must contain at least

$$jL + 1$$

occurrences of action P .

To observe any coordinate of $W^{(j)}$ through the reversed Q -stream, it must contain at least

$$n - jL + 1$$

occurrences of action Q .

Seeing both would therefore require at least

$$(jL + 1) + (n - jL + 1) = n + 2$$

actions. This is impossible within horizon n . □

The correlation has literally been placed beyond the causal anti-diagonal.

18.2 Exactness for every adaptive policy

Proposition 18.2 (Exact causal validity). *The anti-diagonal construction is an exact horizon- n C3-C causal realization of the memoryless channel $\{P, Q\}$.*

Proof. Fix any deterministic action word of length at most n .

Every independent component of the seed contributes to the realized path through at most one of its two coupled marginal blocks, by the anti-diagonal lemma. The exposed coordinates from that marginal therefore have exactly their product law $P^{\otimes L}$ or $Q^{\otimes L}$.

Different coupled components were drawn independently. Hence all outputs exposed along the fixed action word have exactly the product law required by the memoryless controlled channel.

Now fix a deterministic adaptive policy and an output sequence. That output sequence determines one action word. The fixed-word statement applies to that branch, giving exact adaptive validity. $\square$

18.3 Entropy of the causal seed

Since the components are independent,

$$H(S_n) = Lh_P + Lh_Q + (m-1)H(J_L). \quad (18.5)$$

Using (??),

$$H(S_n) \leq Lh_P + Lh_Q + (m-1) \left(Lh_\star + A\sqrt{L} + c_0 \right).$$

Let

$$h_{\min} = \min\{h_P, h_Q\}.$$

Since $n = mL$,

$$\boxed{H(S_n) \leq nh_\star + Lh_{\min} + (m-1) \left(A\sqrt{L} + c_0 \right)}. \quad (18.6)$$

Dividing by $n = mL$,

$$\boxed{\frac{H(S_n)}{n} \leq h_\star + \frac{h_{\min}}{m} + \frac{A}{\sqrt{L}} + \frac{c_0}{L}}. \quad (18.7)$$

19 The causal Shannon-floor theorem

We can now close the two-action causal rate problem.

Theorem 19.1 (Causal Shannon floor for two actions). *Let P, Q be arbitrary finite probability distributions. For the finite-horizon history-indexed causal contract C3-C,*

$$\boxed{C_\infty^{\text{causal}}(P, Q) = \max\{H(P), H(Q)\}}. \quad (19.1)$$

Proof. Set

$$h_\star = \max\{H(P), H(Q)\}.$$

Lower bound. The fixed policy that always selects an entropy-maximizing action produces an n -step trajectory with entropy

$$nh_\star.$$

That trajectory is a deterministic function of the complete causal seed. Hence

$$C_n^{\text{causal}} \geq nh_\star$$

for every n , and therefore

$$C_\infty^{\text{causal}} \geq h_\star.$$

Upper bound. For horizons

$$n = mL,$$

the anti-diagonal construction gives

$$\frac{C_{mL}^{\text{causal}}}{mL} \leq h_{\star} + \frac{h_{\min}}{m} + \frac{A}{\sqrt{L}} + \frac{c_0}{L}.$$

Let

$$m \rightarrow \infty, \quad L \rightarrow \infty.$$

All correction terms vanish, so

$$C_{\infty}^{\text{causal}} \leq h_{\star}.$$

Combine both inequalities. □

19.1 A finite-horizon quantitative corollary

The construction yields more than the limiting rate.

Corollary 19.2 (Sublinear causal residual). *For fixed finite P, Q ,*

$$\boxed{C_n^{\text{causal}} \leq nh_{\star} + O(n^{2/3}).} \tag{19.2}$$

In the equal-entropy case,

$$\boxed{E_n = O(n^{2/3}).} \tag{19.3}$$

Proof. Choose block length

$$L \asymp n^{2/3}$$

and

$$m \asymp n^{1/3}.$$

Then the three excess terms in (??) satisfy

$$Lh_{\min} = O(n^{2/3}),$$

$$mA\sqrt{L} = O(n^{2/3}),$$

and

$$mc_0 = O(n^{1/3}).$$

If $mL < n$, realize the remaining fewer than L positions by any fixed exact finite causal construction. Its additional cost is $O(L) = O(n^{2/3})$. □

Thus finite residual hidden-tree entropy may grow, but it need not be extensive.

20 The crash test is now completely separated

For the concrete pair,

$$H(P) = H(Q) = 1.5.$$

The causal Shannon-floor theorem gives exactly

$$\boxed{C_{\infty}^{\text{causal}}(P, Q) = 1.5.} \tag{20.1}$$

The synchronous information-spectrum theorem gives

$$C_{\infty}^{\text{syn}}(P, Q) \geq 1.728465468745531. \quad (20.2)$$

The explicit synchronous witness gives

$$C_{\infty}^{\text{syn}}(P, Q) \leq 1.939147788079282. \quad (20.3)$$

Therefore

$$\boxed{1.5 = C_{\infty}^{\text{causal}} < 1.728465468745531 \leq C_{\infty}^{\text{syn}} \leq 1.939147788079282 < C_1 = 2.053918585607260.} \quad (20.4)$$

In particular,

$$\boxed{C_{\infty}^{\text{syn}} - C_{\infty}^{\text{causal}} \geq 0.228465468745531.} \quad (20.5)$$

This is a strict asymptotic separation between C3-S and C3-C.

Hidden correlations alone were already known to reduce the synchronous one-shot bill. The new causal construction shows something stronger: history-indexed counterfactual dependence can move the entire *rate* surcharge behind the causal horizon.

21 Finite non-attainment versus asymptotic attainment

The causal result has a useful non-attainment structure.

For the crash test,

$$E_1 = 0.553918585607260 \dots > 0.$$

Since E_n is nondecreasing,

$$E_n \geq E_1 \quad \forall n. \quad (21.1)$$

Hence

$$\boxed{C_n^{\text{causal}} > 1.5n \quad \text{for every finite } n.} \quad (21.2)$$

The depth-two rigidity theorem strengthens this to

$$E_2 > E_1. \quad (21.3)$$

Nevertheless, the anti-diagonal construction proves

$$E_n = O(n^{2/3}),$$

so

$$\frac{E_n}{n} \rightarrow 0. \quad (21.4)$$

Thus

$$\boxed{\text{the Shannon floor is never reached at finite horizon, but is reached as an asymptotic rate.}} \quad (21.5)$$

This is precisely the distinction that the earlier finite-reset obstruction could not decide.

22 What resource does the synchronous tax price?

The hierarchy for the crash-test pair is now

$$\begin{aligned}
&\text{chosen path} && : & 1.5, \\
&\text{general causal tree C3-C} && : & 1.5, \\
&\text{synchronous stable-event C3-S} && : & \geq 1.7284654687, \\
&\text{one-step independent-table benchmark} && : & 2.0539185856.
\end{aligned} \tag{22.1}$$

The causal and synchronous contracts both reproduce every adaptive observable policy exactly. Their difference is not observable-path correctness.

The difference is the identity relation imposed on unrealized events.

Under C3-S,

$$F_t(a)$$

is the same latent coordinate regardless of which earlier history would have led to time t .

Under C3-C,

$$F_t(h_{t-1}, a)$$

may be a different latent event for different histories.

The anti-diagonal construction exploits precisely this freedom. It stores dependence between futures that cannot jointly lie on any single admissible horizon- n path.

For the present two-action example, the positive synchronous asymptotic tax therefore prices

stable cross-history event identity,

not causal counterfactual coherence by itself.

23 Why this is not an ontological theorem

The mathematical conclusion is contract-dependent.

It does not imply that observed stochasticity is ontic. It does not identify a fundamental substrate. It does not disprove simulation. It does not show that a physical realization must literally contain a Shannon-coded response table.

It says:

If one requires a particular class of exact latent realizations, then the minimum information source compatible with that class obeys the corresponding lower and upper bounds.

Different realization classes may exchange resources between:

$$(M, R, \Pi, B, I), \tag{23.1}$$

where

- M is persistent memory;
- R is fresh innovation;
- Π is precision or description complexity;
- B is counterfactual breadth;
- I is the strength of stable cross-history event identity.

Exact real-valued state can hide arbitrarily much discrete information in its precision. Extensive initial preload can substitute for later innovation. Weakening counterfactual breadth can remove branching obligations. Weakening stable identity from C3-S to C3-C can remove an entire asymptotic rate surcharge.

The resource theorem therefore constrains implementations. It does not identify ontology.

24 Finite-horizon rate versus one infinite substrate

The causal Shannon-floor theorem concerns

$$C_n^{\text{causal}}$$

optimized separately at every finite horizon and the subadditive rate

$$C_\infty^{\text{causal}} = \inf_n \frac{C_n^{\text{causal}}}{n}.$$

The anti-diagonal source used at horizon n depends on the location of that horizon. Its reversed block geometry is therefore not automatically projectively consistent as n changes.

Consequently, the theorem does *not* by itself construct one random infinite causal tree whose restrictions to all finite horizons are simultaneously entropy-optimal.

This motivates a stronger accounting standard.

Definition 24.1 (Projectively consistent causal substrate). *An infinite causal substrate is a probability law on one infinite history-indexed response tree whose restriction to every finite depth is an exact causal realization.*

One may then ask whether an appropriate entropy-rate notion for a single such infinite substrate still equals $h_{\max}$.

This is a different problem. The present anti-diagonal theorem does not answer it.

25 What is proved

Under the synchronous stable-event contract C3-S:

- transversal correctness is equivalent to exactness for every adaptive nonanticipative policy within the synchronous representation;
- C_n^{syn} is subadditive and has a well-defined asymptotic rate;
- the transversal marginal polytope has rank

$$d_{\text{obs}}^n;$$

- zero-conditional-mean interaction directions generate hidden cross-time synergy;
- the crash test has an explicit 23-atom witness with

$$C_2^{\text{syn}} \leq 3.878295576158564;$$

- hidden synergy has the exact variational identity (??);
- the scalar probability minimum in the synchronous converse is exactly the reciprocal-scale $k = 0$ kernel overlap;

- for two actions,

$$C_{\infty}^{\text{syn}} \geq \frac{1}{2} (\mathcal{H}(P) + \mathcal{H}(Q) + W_1(\mu_P, \mu_Q));$$

- the converse extends to arbitrary finite action families through a multi-marginal quantile envelope;
- for the crash-test pair,

$$1.728465468745531 \leq C_{\infty}^{\text{syn}} \leq 1.939147788079282.$$

Under the general history-indexed causal contract C3-C:

- causal trees admit the action-tree quotient (??);
- C_n^{causal} is subadditive and obeys the causal prefix lower bound;
- for equal-entropy actions,

$$C_n^{\text{causal}} = nh + E_n,$$

where E_n is nonnegative, nondecreasing, and subadditive;

- E_n is the minimum residual hidden-tree entropy after one realized policy trajectory is revealed;
- an exact Bellman–MEC recursion holds;
- for the crash test,

$$E_2 > E_1 > 0;$$

- for every finite two-action channel,

$$C_{\infty}^{\text{causal}}(P, Q) = \max\{\mathcal{H}(P), \mathcal{H}(Q)\};$$

- quantitatively,

$$C_n^{\text{causal}} \leq nh_{\max} + O(n^{2/3});$$

- for the crash test,

$$C_{\infty}^{\text{causal}} = 1.5 < C_{\infty}^{\text{syn}}.$$

26 What is not proved

The following statements are *not* established here.

- The explicit 23-atom synchronous witness is not proved to minimize C_2^{syn} .
- The exact value of C_{∞}^{syn} for the crash-test pair remains unknown.
- No theorem is given here that the $O(n^{2/3})$ causal excess is optimal. The true finite-horizon residual could be bounded, logarithmic, or obey another sublinear law.
- The two-action causal Shannon-floor theorem is not claimed here for arbitrary numbers of actions.
- The finite-horizon construction does not automatically produce one projectively consistent infinite causal substrate.
- No claim of historical novelty is made for MEC, information-spectrum methods, quantile couplings, hidden synergy, marginal-polytope geometry, or the underlying coding ideas.
- No entropy inequality here identifies the physical substrate, disproves simulation, or decides whether observable randomness is fundamental.

27 Open problems after Version 4

Open Problem 27.1 (Exact synchronous asymptotic rate). *For the crash-test pair, determine*

$$C_{\infty}^{\text{syn}} \in [1.728465468745531, 1.939147788079282].$$

Is the information-spectrum transport converse sharp?

Open Problem 27.2 (Exact synchronous two-step entropy). *Determine*

$$C_2^{\text{syn}}$$

exactly. The current certified bracket is

$$3.553918585607260 \leq C_2^{\text{syn}} \leq 3.878295576158564.$$

Open Problem 27.3 (Finite causal residual). *Determine the true asymptotic order of*

$$E_n = C_n^{\text{causal}} - nh$$

for the crash-test pair.

Version 4 proves

$$E_1 > 0, \quad E_2 > E_1, \quad E_n = O(n^{2/3}),$$

but does not determine whether E_n is bounded, logarithmic, or follows another sublinear law.

Open Problem 27.4 (Multi-action causal Shannon floor). *For*

$$|\mathcal{A}| > 2,$$

does

$$C_{\infty}^{\text{causal}}(P_a : a \in \mathcal{A}) = \max_a H(P_a)$$

still hold?

The two-stream anti-diagonal construction does not immediately provide a multi-action proof.

Open Problem 27.5 (Projectively consistent infinite causal substrate). *Can the finite-horizon causal Shannon-floor construction be replaced by one infinite projectively consistent random response tree with the same asymptotic resource rate?*

Open Problem 27.6 (Sharper anti-diagonal coding). *Can the boundary term in the causal construction be reduced sufficiently to improve*

$$O(n^{2/3})$$

to

$$O(\sqrt{n}), \quad O(\log n), \quad \text{or } O(1)?$$

Open Problem 27.7 (Multi-action synchronous information-spectrum geometry). *Characterize sharp synchronous lower bounds for large action families beyond the expected-maximum quantile envelope.*

Open Problem 27.8 (Higher-order reciprocal-scale kernels). *Does any $k > 0$ member of the compact-support kernel hierarchy combine with simultaneous transversal consistency to yield a converse stronger than the $k = 0$ information-spectrum bound?*

Open Problem 27.9 (Breadth-rate geometry). *For families*

$$\mathcal{G}_n \subseteq \mathcal{A}^n,$$

characterize the asymptotic source rate by intrinsic covering or coding geometry. Cardinality alone is insufficient.

28 Substrate skepticism after the mathematics

The original question asked who pays for exact stochastic counterfactual structure.

Version 4 gives a more differentiated answer.

A single realized path pays the ordinary Shannon bill.

A general finite-horizon causal response tree with two actions also pays, asymptotically, only the largest observable Shannon rate:

$$C_{\infty}^{\text{causal}} = h_{\max}.$$

A synchronous stable-event table can cost strictly more, because it requires mutually exclusive histories to share the same latent event identity. For the crash-test pair,

$$C_{\infty}^{\text{syn}} \geq 1.7284654687 > 1.5.$$

Thus the extra synchronous bill cannot be attributed merely to the existence of unrealized alternatives. It is tied to a stronger compatibility requirement among them.

The strongest statement supported here is therefore:

We do not infer what reality is made of.
 We can price explicit notions of counterfactual consistency.
 For a finite two-action memoryless channel, unrestricted history-indexed causal coherence reaches the observable Shannon floor as an asymptotic finite-horizon rate.
 If the same time/action event must instead retain a stable synchronous identity across mutually exclusive histories, a mismatch of self-information spectra can leave a strictly positive asymptotic source bill.

The ontological question remains unanswered.

The accounting question has become more precise.

29 Conclusion

The attack changed its target repeatedly.

The one-shot MEC surcharge was real but too large to be interpreted as a per-step tax.

Chosen-path randomness recycling showed that adaptive sampling alone can approach Shannon.

The strong-reset obstruction showed that exact finite recycling at the Shannon floor can fail when probability spectra differ.

Removing reset independence exposed hidden synergy: mutually exclusive futures can carry dependence invisible to every individual selectable coordinate.

The synchronous information-spectrum converse then showed that this synergy cannot eliminate the whole bill when the same latent time/action event is required to remain identical across histories.

The remaining question was whether a fully history-indexed causal tree still paid a positive asymptotic tax.

For two actions, Version 4 answers that question:

$$C_{\infty}^{\text{causal}}(P, Q) = \max\{H(P), H(Q)\}.$$

The mechanism is anti-diagonal hidden dependence. Long product blocks are coupled almost at the Shannon floor, but the coupled pieces are placed so that no admissible finite-horizon history can observe both sides of the same correlation.

For the concrete equal-entropy crash test,

$$1.5 = C_{\infty}^{\text{causal}} < 1.728465468745531 \leq C_{\infty}^{\text{syn}} \leq 1.939147788079282 < 2.053918585607260 = C_1.$$

Thus the one-shot bill was too large. The Shannon bill was exactly right for unrestricted causal coherence. The positive asymptotic remainder belongs to the stronger synchronous identity contract.

A history says what happened.

A causal rule says what would happen after every history.

A synchronous rule demands something more: that mutually exclusive histories agree on what counts as *the same unrealized event*.

Under the present accounting standard, that additional identity has a price.

Who pays depends on what must remain the same.

A The explicit 23-atom synchronous witness

Write each atom as

$$(x_1, z_1, x_2, z_2) \in \{0, 1, 2\}^4,$$

where x_t is the P -coordinate and z_t is the Q -coordinate at time t .

The nonzero probabilities of the witness $\tilde{G}$ are:

(x_1, z_1, x_2, z_2)	$\tilde{G}(x_1, z_1, x_2, z_2)$
$(0, 0, 0, 0)$	$\frac{(2q-1)(12q-5)}{8}$
$(0, 0, 0, 2)$	$-q(2q-1)$
$(0, 0, 2, 1)$	$\frac{q}{4}$
$(0, 1, 0, 0)$	q^2
$(0, 1, 0, 2)$	$-\frac{(2q-1)(8q-3)}{8}$
$(0, 1, 1, 1)$	$\frac{q}{4}$
$(0, 2, 1, 0)$	$-\frac{2q-1}{8}$
$(0, 2, 2, 0)$	$-\frac{(2q-1)(8q-3)}{8}$
$(0, 2, 2, 2)$	$\frac{(2q-1)(16q-7)}{8}$
$(1, 0, 0, 0)$	$\frac{q}{4}$
$(1, 0, 1, 1)$	$\frac{4q-1}{16}$
$(1, 1, 0, 2)$	$-\frac{2q-1}{8}$
$(1, 1, 2, 1)$	$\frac{1}{16}$
$(1, 2, 1, 2)$	$-\frac{2q-1}{8}$
$(2, 0, 0, 0)$	$-\frac{32q^2-40q+11}{16}$
$(2, 0, 0, 1)$	$\frac{(4q-1)^2}{16}$
$(2, 0, 1, 0)$	$\frac{1}{16}$
$(2, 1, 0, 1)$	$\frac{16q^2-4q-1}{12q-5}$
$(2, 1, 0, 2)$	$-\frac{16}{4q-1}$
$(2, 1, 2, 2)$	$\frac{16}{(2q-1)^2}$
$(2, 2, 0, 0)$	$\frac{2}{-q(2q-1)}$
$(2, 2, 0, 1)$	$-\frac{2q-1}{8}$
$(2, 2, 2, 0)$	$-\frac{2q-1}{8}$

(A.1)

The entries sum symbolically to one.

For every

$$a, b \in \{P, Q\},$$

summing over the two unselected coordinates gives exactly

$$P_a \otimes P_b.$$

At the equal-entropy root q from (??), all displayed atoms are strictly positive.

B One-step MEC spectral rigidity

For completeness, the five vertex-spectrum types of the one-step 3×3 transportation polytope can be taken as

$$\begin{aligned}\lambda_\star &= \left(q, \frac{1}{4}, 1 - 2q, \frac{1}{2} - q, 2q - \frac{3}{4}\right), \\ \lambda_A &= \left(\frac{1}{4}, \frac{1}{4}, 1 - 2q, q - \frac{1}{4}, q - \frac{1}{4}\right), \\ \lambda_B &= \left(\frac{3}{4} - q, \frac{1}{4}, 1 - 2q, q - \frac{1}{4}, 2q - \frac{3}{4}\right), \\ \lambda_C &= \left(2q - \frac{1}{2}, \frac{1}{4}, 1 - 2q, q - \frac{1}{4}, \frac{1}{2} - q\right),\end{aligned}$$

and

$$\lambda_D = \left(q, \frac{1}{4}, q - \frac{1}{4}, \frac{1}{2} - q, \frac{1}{2} - q\right).$$

On

$$\frac{3}{8} < q < \frac{5}{12},$$

the cumulative-sum differences $\lambda_\star - \lambda_J$ are:

	1	2	3	4	5
A	$q - \frac{1}{4}$	$q - \frac{1}{4}$	$q - \frac{1}{4}$	$\frac{1}{2} - q$	0
B	$2q - \frac{3}{4}$	$2q - \frac{3}{4}$	$2q - \frac{3}{4}$	0	0
C	$\frac{1}{2} - q$	$\frac{1}{2} - q$	$\frac{1}{2} - q$	$\frac{5}{4} - 3q$	0
D	0	0	$\frac{5}{4} - 3q$	$\frac{5}{4} - 3q$	0

(B.1)

Every displayed nonzero entry is positive. Therefore

$$\lambda_\star \succ \lambda_J \quad (J = A, B, C, D).$$

Strict Schur concavity of Shannon entropy gives the uniqueness of the minimum atom spectrum used in the depth-two rigidity argument.

C Reproducibility ledger

For the crash-test pair, the numerical values used in the synchronous part are

$$q = 0.41003242818198557608\dots,$$

$$C_1 = 2.0539185856072597\dots,$$

$$H(\tilde{G}) = 3.878295576158564\dots,$$

$$\frac{1}{2}H(\tilde{G}) = 1.939147788079282\dots,$$

$$W_1(\mu_P, \mu_Q) = 0.4569309374910615\dots,$$

and

$$1.5 + \frac{1}{2}W_1 = 1.7284654687455308 \dots$$

The companion Version 3 verification script symbolically checks:

1. the equal-entropy root equation;
2. normalization of the 23-atom witness;
3. all four two-step transversal identities;
4. positivity of the displayed atoms;
5. the hidden mutual-information identities;
6. the $k = 0$ reciprocal-scale overlap identity.

The new causal Shannon-floor theorem is analytic. It does not depend on a numerical greedy causal recursion.

D Literature map

Topic	Status used here	Representative neighbor
Minimum-entropy coupling	Known	Kovačević–Stanojević–Šenk
Information-spectrum MEC converse	Known	Shkel–Yadav
Approximate multi-marginal MEC	Known	Li
Greedy/profile MEC bound	Known	Compton et al.
Small sample spaces under constraints	Known	Koller–Megiddo
Exact common-information rate	Known	Kumar–Li–El Gamal
Exact channel synthesis	Known	Yu–Tan
Joint entropy under structured independence	Known	Gavinsky–Pudlák
Marginal polytopes and codes	Known	Kahle–Wenzel–Ay
Chosen-path randomness recycling	Known	Draper–Saad; Draper–Harris–Saad
Mixed versus behavioral strategies	Known	Kuhn; Main–Randour
Causal transport	Known	Cont–Lim
Synergistic disclosure	Known	Rassouli–Rosas–Gündüz
Stable event-keyed randomness	Neighboring issue	Buffalo–Pearson–Klein
Compact-support overlap kernel	Earlier author note	Mikulík
Synchronous entropy under all complete transversals	Not located in this formulation	Compare structured independence and MEC
Synchronous quantile-envelope converse	Not located in this formulation	Compare information-spectrum MEC converses
Two-action finite-horizon causal Shannon-floor theorem	Not located in this formulation	Proof here uses block MEC profile bound
Infinite projectively consistent causal rate	Open here	Compare causal transport and finitary coding

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